UFUG 2106 : Discrete Mathematics

香港科技大学(广州)
THE HONG KONG
UNIVERSITY OF SCIENCE AND
TECHNOLOGY (GUANGZHOU)

Course team

- Instructor: Shiju Lin
  - Assistant Professor, Microelectronics Thrust
  - Office hours: Tuesday 19:00—21:00 (location: E1 309)
  - Email: shijulin@hkust-gz.edu.cn

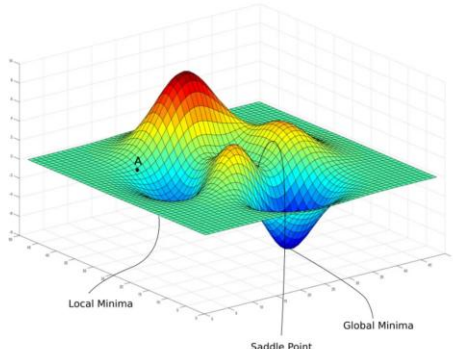
- Teaching assistants:
  - Yewei Wang (ywang056@connect.hkust-gz.edu.cn)
  - Zhuofan Lin (zlin037@connect.hkust-gz.edu.cn)

What is discrete mathematics?



Discrete Mathematics

- Discrete mathematics is a branch of mathematics that is concerned with "**discrete**" mathematical structures instead of "**continuous**"
- Continuous mathematics:
  - Real numbers
  - Calculus
  - Euclidean geometry
  - Etc.
- Discrete mathematics:
  - Integers, number theory
  - Logic
  - Set theory
  - Combinatorics
  - Graph theory
  - Etc.



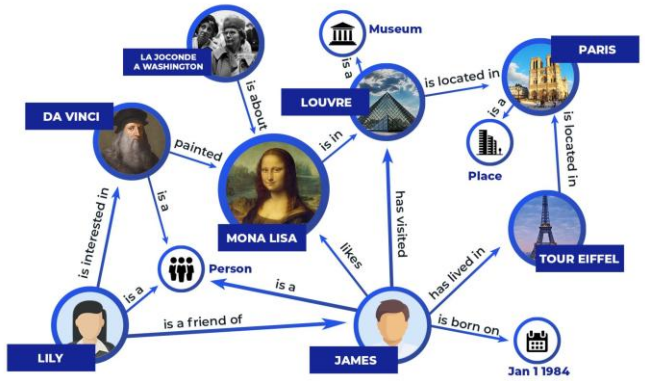


Image source: https://community.atlassian.com/t5/Confluence-questions/Knowledge-graph/qaq-p/1565284

Why does it matter?

- Digital computers are built as a discrete machine: 0 and 1
  - How to use 0s and 1s to solve real-world problems?
- A mathematical foundation for
  - Computer science topics: data structures, algorithms, relational database, compilers, cryptography, artificial intelligence, etc.
  - Hardware design topics: logic synthesis, placement, routing, etc.
- Core knowledge in computer science curricular
  - Recommended by both IEEE (Institute of Electrical and Electronics Engineers) and ACM (Association for Computing Machinery)

What you are expected to learn (Intended Learning Outcomes)

By the end of this course, students should be able to:

1. Understand the formulation of common problems in several areas of discrete mathematics, including logic and proof, combinatorics, recursion and recurrences, and graph theory
2. Understand and analyze the key algorithms in the above areas of discrete mathematics
3. Plan and select appropriate concepts and algorithms in the above areas to solve real-world problems

What you'll be doing every week

- Lectures L01: Monday 01:30PM - 04:20PM, E1 201
- Tutorials T01: Friday 09:00AM - 09:50AM, E1 201
- Mid-term test: around the 7th week

Course Schedule (tentative)

Week	Topics	Weekly ILOs
1	Speaking mathematically	ILO1
2	The logic of compound statements	ILO1, ILO2, ILO3
3	The logic of quantified statements	ILO1, ILO2, ILO3
4	Number theory and methods of proof	ILO1, ILO2, ILO3
5	Sequences, mathematical induction, and recursion	ILO1, ILO2, ILO3
6	Set theory	ILO1, ILO2, ILO3
7	Mid-term test; Properties of functions	ILO1, ILO2, ILO3
8	Properties of relations	ILO1, ILO2, ILO3
9	Counting and probability	ILO1, ILO2, ILO3
10	Theory of graphs and trees	ILO1, ILO2, ILO3
11	Analysis of algorithm efficiency	ILO1, ILO2, ILO3
12	Regular expressions and finite-state automata	ILO1, ILO2, ILO3
13	Finite-state automata; course review	ILO1, ILO2, ILO3

Recommended textbooks

- Main textbook:
  - "**Discrete Mathematics with Applications**", 5th Edition, Susanna S. Epp
  - Available at HKUST(GZ) library (online)

- Check the following reference book for more questions to practice
  - "Discrete Mathematics and Its Applications", 8th Edition, Kenneth H. Rosen

Prerequisites

- Knowledge of computer science principles and skills
- Familiar with programming
  - There will be a **coding task** to help you practice the knowledge you have learned.

How will I be evaluated?



Overall evaluation scheme

- Continuous Assessment: 30%
  - Written assignments (10%): around one assignment per Chapter
  - Course participation (5%): quizzes
  - Projects (15%): one project
- Mid-term test: 25%
  - Closed book
- Final exam: 45%
  - Closed book

Late Submission Policy

- 1-3 Days Late: 10% of the total possible points will be deducted from the score.
- 4-7 Days Late: 20% of the total possible points will be deducted.
- More than 7 Days Late: The assignment will not be accepted and will receive a score of zero.
- Extensions:
  - Extensions will only be granted in exceptional circumstances, such as documented medical emergencies or other serious situations. Students must contact the instructor as soon as possible to request an extension.
  - If there are technical difficulties with the submission platform, students must email the instructor immediately with evidence of the issue (such as screenshots) and submit the assignment as soon as the issue is resolved.

Anything more?



What do we expect

- Before attending each lecture:
  - Read the textbook (around 30 pages)
- During the lecture:
  - Learn the concepts and examples (to enhance your understanding)
  - Work on quizzes (as a self-evaluation)
- After the lecture:
  - Work on the assignments independently (and review the textbook when necessary)
  - Work on the project (apply the knowledge and skills on real problems)
- Tutorials by Teaching Assistants:
  - More challenging examples and review of common mistakes in exercises
- Office hours:
  - Face-to-face discussions

Policy on the use of AI tools

- Students **are encouraged to** use generative artificial intelligence (GAI) tools for self-studying and exploring the topics of discrete mathematics.

- However, students are **not allowed** to use GAI tools to complete assessment tasks in this course.

Academic Integrity

- Academic integrity
  - Students are expected to adhere to the university's academic integrity policy. Students are expected to uphold HKUST(GZ)'s Academic Honor Code and to maintain the highest standards of academic integrity. The University has zero tolerance of academic misconduct. Please refer to Regulations for Academic Integrity and Student Conduct for the University's definition of plagiarism and ways to avoid cheating and plagiarism.
  - https://vptlo.hkust-gz.edu.cn/ugeducation/#/ugstudenthandbook
- How to avoid plagiarism and copying?
  - https://www.hkust-gz.edu.cn/academics/academic-quality-manual/roles-and-responsibilities-students/how-to-avoid-plagiarism-and-copying/
- Example case:
  - Student A borrowed the completed assignment from Student B for whatever reason. Student A then partially copied B's solutions and submitted his assignment.
  - The software Turnitin detected that the submitted assignments from A and B have significant similarities. Course instructors investigated the case.
  - Both A and B will receive zero marks on this assignment. The case will be reported to Thrust for further action.